Modelling the Olfactory Bulb

Technical Milestone Report

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Summary

This report presents a mathematical model of the mouse olfactory bulb for investigating homeostatic plasticity. The model includes all major neuron types and their main interactions. Analysis reveals substantial parameter degeneracy, with multiple configurations producing the same output for a given input. Stability analysis has confirmed robustness to small perturbations, and Hessian-based analysis revealed directions in parameter space that preserve the output, a property required to reproduce experimentally observed homeostatic plasticity. Results suggest homeostatic compensation can emerge from distributed synaptic adjustments, and future work will focus on constraining solutions to biologically plausible synaptic changes.

1 Introduction

Biological systems need to maintain stable internal conditions despite disturbances in the external environment, a process known as homeostasis. Common examples include the regulation of body temperature and blood glucose concentration. Homeostasis can be viewed as sophisticated feedback control loops that keep key variables within healthy and functional ranges.

Neural circuits require similar regulation, as insufficient activity impairs information transmission, while excessive activity can lead to disorders such as epilepsy or chronic pain.[1,2] To maintain activity within a functional range, neurons rely on homeostatic plasticity, a mechanism that adapts synaptic strength and intrinsic properties to stabilise overall activity.

Homeostatic plasticity is ubiquitous throughout the nervous system.[2–5] However, the complexity of these circuits and their limited accessibility make experimental investigation very challenging.

Here, we use computational methods to complement experimental studies and develop a model of the mouse olfactory bulb (OB), a well-characterised system exhibiting homeostatic plasticity. Its compact, accessible network and experimental data from the Galliano Laboratory,[6] make this an ideal system for the study. The ultimate aim is to identify mechanisms underlying homeostasis in the OB, generate testable predictions, and guide future experiments.

2 Olfactory Bulb (OB)

2.1 OB: Sensory Deprivation

Sensory deprivation has been shown to induce homeostatic plasticity in the olfactory bulb (OB) in mice.[5] The experimental procedure is described only briefly here. When an odour is presented to the mouse, neural signals propagate from olfactory sensory neurons (OSNs) through the OB to the cerebral cortex. The OB is ipsilateral, so blocking one nostril affects only the OB on the same side. Sensory deprivation can therefore easily be studied by simply placing a nose plug in one of the nostrils (allowing the mouse to breath through the other one). Experimental results show that 24-hour deprivation causes dopaminergic axonic neurons to shrink in axonal length.[5]

Interestingly, when the nose plug is removed, the OB output returns to its original level despite the structural changes undergone during the sensory deprivation period. This finding is striking, comparable to a limb retaining its strength after cast removal despite muscle weakening from a period of immobilisation. This remarkable observation suggests that compensatory mechanisms, yet to be identified, must occur elsewhere in the OB.

2.2 OB: Key Neuron Types

To investigate possible mechanisms of homeostatic plasticity, the first stage of the project involved developing a mathematical model incorporating its principal neuron types and interactions.[7]

Each odour has a characteristic chemical composition, and its molecules bind selectively to OSNs, producing a unique pattern of activity in the OB (Fig. 1). Interestingly, OSNs expressing the same receptor type converge onto the same glomerulus, a region within the OB where the first stage of neural processing of smells occurs. The glomerulus consists of a cluster of synapses and it is here where the axon terminals of the OSNs release their neurotransmitter. Beside OSNs, many additional interneurons are at play in the glomerulus.

For modelling purposes, we group interneurons into functional groups:

- Anaxonic interneurons (blue) provide self inhibition within a glomerulus, preventing over-excitation.
- Axonic interneurons (green) provide long-range lateral inhibition, receiving input from one glomerulus and inhibiting neighbouring glomeruli.
- External tufted cells (ETCs, red) relay signals from glomeruli to mitral cells. The mouse brain contains two olfactory bulbs, one per hemisphere, each with two "sister" glomeruli per receptor type. ETCs connect the sisters and transmit signals to the corresponding mitral cell. In the model, sister glomeruli are merged, yielding a single ETC per glomerulus.
- Mitral cells (grey) integrate signals from the glomeruli and form the primary output to the cortex.
- Granule cells (black) provide feedback inhibition to mitral cells. As they receive cortical input, they are thought to support learning processes.

For convenience, we refer to anaxonic interneurons as blue (b), axonic interneurons as green (g), and ETCs as red (r), following the colour scheme in Fig. 1.

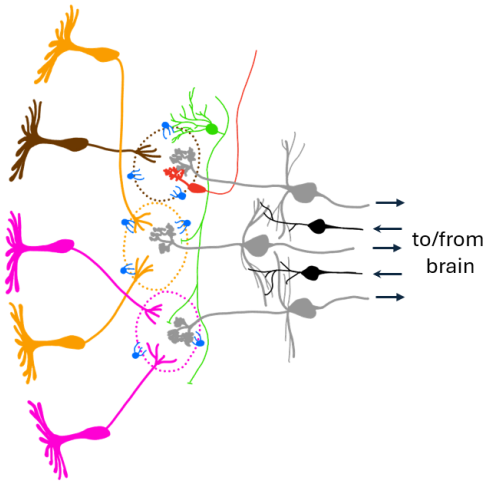


Figure 1: Simplified OB circuitry: OSNs (pink/orange/brown), axonic interneurons (green), anaxonic interneurons (blue), ETCs (red), mitral cells (grey), granule cells (black).

3 Mathematical Model

3.1 Model: Neuron

Fig. 2 illustrates the way the interaction between two neurons is computed in the model. The first neuron (yellow) receives a stimulus x and generates an output z through a function σ_z, such that $z = \sigma_z(x)$. A fraction of this output, $w_z\ z$ (with $0 < w_z < 1$) serves as an input to the second neuron (purple), which applies its own function σ_y to generate the output $y = \sigma(w_z\ z)$. In the model, input/output variables represent steady-state firing rates, the weights w_i represent the strength of the synaptic links and σ_j the response of neurons to stimuli.

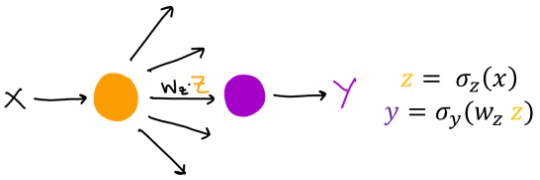


Figure 2: Two-neuron circuit model: x, z, y are firing rates and w synaptic link strength.

Neuronal responses are modelled with the Naka–Rushton function,[8] capturing their nonlinear saturation behaviour while being computationally simple and differentiable:

$$\sigma_j(x) = \frac{x^{n_j}}{x^{n_j} + k_j^{n_j}} \qquad (1)$$

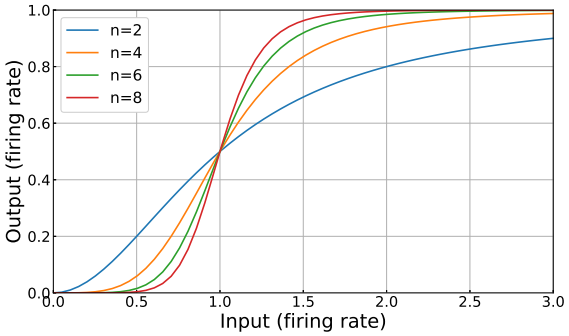


Figure 3: Naka-Rushton functions (Eq. (2)).

Here, k_j is the semi-saturation constant and n_j the Hill coefficient, which control when and how sharply the output transitions from 0 to 1. For computational efficiency, all inputs can be scaled $x' = x/k_j$:

$$\sigma_j(x') = \frac{x'^{n_j}}{x'^{n_j} + 1} \tag{2}$$

The scaling factor $(1/k_j)$ can then be absorbed by the input weight w by simply allowing for weights to take any positive value. Fig. 3 shows Eq. (2) for various values of n. For the set of results presented here, we fix $n = 4$.

3.2 Model: Olfactory Bulb

With the key neuron types and connectivity identified (Section 2.2), we construct a mathematical model of the OB using the neuronal model from Section 3.1. The model is built on several key assumptions. First, neurons are grouped by "type" and represented by a single representative cell for each group, enabling the model to capture the essential behaviour of each cell population without simulating every individual neuron. The model includes two glomeruli to account for cross-glomerular inhibition by the axonic neurons (green in Fig. 4) and anaxonic neurons (blue) are assumed to inhibit only their own glomerulus.

In addition, since we are interested in studying homeostatic plasticity and comparing long-term end states rather than transient fluctuations, the model neglects temporal dynamics and considers only steady-state firing rates.

The steady-state firing rates of blue (b), red (r), green (g), and mitral (y) neurons are then given by:

$$b = \sigma\left(w_{\text{osn},b}\,\alpha x\right) \tag{3}$$

$$r = \sigma\left(w_{\text{osn},r}\,\alpha x - w_{b,r}\,b - \mathbf{W}_{g,r}\,g\right) \tag{4}$$

$$g = \sigma\left(w_{\text{osn},g}\,\alpha x - w_{b,g}\,b + w_{r,g}\,r\right) \tag{5}$$

$$y = \sigma(w_{\text{osn},y}\,\alpha x - w_{b,y}\,b - \mathbf{W}_{g,y}\,g + w_{r,y}\,r - w_{gr,y}\,gr) \tag{6}$$

$$gr = \sigma\left(w_{\text{cor,gr}}\,\text{cortex} + w_{y,gr}\,y\right) \tag{7}$$

where σ is the Naka-Rushton function (Eq. (2)), $w_{i,j}$ is the weight that accounts for the synaptic strength between neurons i and j and the scaling factor $1/k_j$ due to the semi-saturation constant of neuron j (see Section 3.1), and $\alpha \in [0, 1]$ is a parameter modelling the inhibitory feedback from the blue neurons to the OSNs as at steady state, the OSN output is a fraction α of the original signal. Finally, $\mathbf{W}_{i,j}$ are weight matrices of the form $[0, w_{i_1,j_1}; w_{i_2,j_2}, 0]$, relating neurons i and j for each of the glomeruli.

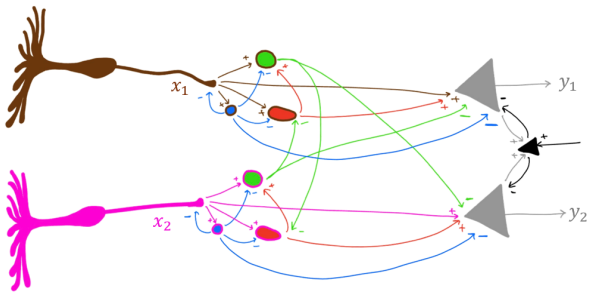


Figure 4: Simplified OB circuit showing the interaction between the main neuron groups.

3.3 Model: Fixed-point Iteration

Equations (3)–(5) form a fixed-point problem since g and r depend on each other. We solve this using Anderson acceleration, which improves convergence by combining information from recent iterations rather than relying only on the previous estimate.[9] Using a history of 5 iterates, a tolerance of 10^{-6}, and a max. of 50 iterations reliably converges to the fixed-point for g.

Similarly, Eq. (6) and (7) form a second fixed-point problem for granule cells. Although granule cells influence mitral cell output, their activity depends on cortical in-

puts. These are poorly characterised and are thought to be more involved in learning than in homeostatic regulation. Due to the lack of experimental data to constrain their parameters and the current understanding that homeostasis is achieved locally, we exclude granule cells from our initial analysis.

3.4 Model: Parameter Initialisation

To model the OB before sensory deprivation, we identify parameters $\theta = \{\omega, \alpha\}$ that produce the target output y for input x while keeping neural firing rates near 0.5 to maximise dynamic range. Biologically, this is reasonable as neurons operating near saturation would convey little information. To achieve this, we start with random parameters and iteratively adjust them until the desired firing conditions and target output y are obtained. This is formulated as an optimisation problem with loss function:

$$L(\theta) = \sum_{i=1}^{m} \big[(r_i - r_{t,i})^2 + (g_i - g_{t,i})^2 + (b_i - b_{t,i})^2 + (y_i - y_{t,i})^2\big] \quad (8)$$

where m is the number of mitral cells (2), r_i, g_i, b_i, y_i the firing rates of the red, green, blue, and mitral neurons, respectively. Target values for each neuron $r_{t,i}$, $g_{t,i}$, $b_{t,i}$, $y_{t,i}$ are all set to 0.5 as discussed above.

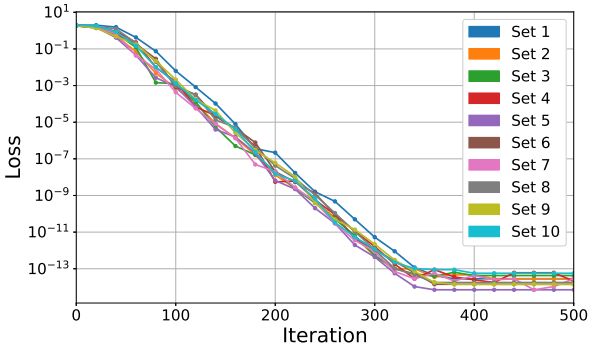


Figure 5: Loss evolution during Adam optimisation for 10 random sets of initial parameters.

We solve this problem using the Adam optimiser, which adaptively adjusts learning rates via exponentially weighted averages of past gradients ($\beta_1 = 0.9$) and squared gradients ($\beta_2 = 0.999$).[10] The algorithm is implemented in JAX/Optax with JIT compilation for efficient automatic differentiation.[11] The optimisation loop enforces biological and numerical constraints by clipping parameters to be positive, restricting α to $[0, 1]$, and setting the diagonal elements of weight matrices $\mathbf{W}$ to zero. Due to system degeneracy, many parameter combinations produce the target output y and neural firing rates. Fig. 5 illustrates convergence for 10 random initialisations.

3.5 Model: Stability Analysis

Because biological neural systems are inherently noisy, any parameter solution identified in Section 3.4 must also be stable for it to be biologically plausible. To assess this, we examine the local stability of the network around the fixed-point solution. We first define the state vector of the firing rates at time t as $s(t) = [r(t), g(t), b(t), y(t)]$ and denote the steady-state solution for a given weight parameter set θ as s^*. By definition:

$$F(s^*, \theta, x) = s^* \quad (9)$$

where F describes the discrete-time dynamics of the neural network, mapping the current firing rates to those in the next time step. Linearising the dynamics around s^*, the evolution of the system after the introduction of a perturbation $e(t)$ is:

$$\begin{aligned} s^* + e(t+1) &= F(s^* + e(t), \theta, x) \\ &\approx F(s^*, \theta, x) + \left.\frac{\partial F}{\partial s}\right|_{s^*} e(t) + \ldots \end{aligned} \quad (10)$$

Using the fixed-point condition (Eq. (9)) in Eq. 10, yields:

$$e(t+1) \approx \mathbf{J} e(t) \quad (11)$$

where J is the Jacobian $\left.\frac{\partial F}{\partial s}\right|_{s=s^*}$. Eq. (11) can be rewritten as a standard continuous-time first order differential equation:

$$\frac{de}{dt} = \frac{1}{\tau}(\mathbf{J} - \mathbf{I})\, e = \mathbf{A}\, e \quad (12)$$

where τ is a time scaling factor and $\mathbf{A}$ the linearised system matrix.

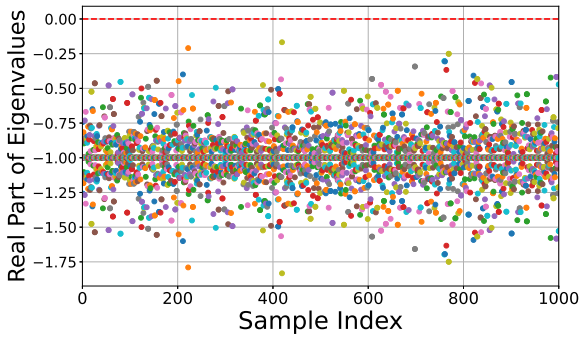


Figure 6: Real parts of eigenvalues of $\mathbf{A}$ for 1000 parameter configurations.

Therefore, the fixed point solution s^* is locally asymptotically stable if all eigenvalues of $\mathbf{A}$ have negative real parts, as that will ensure that perturbations $e(t)$ decay over time. Note that stability (i.e. negative real parts of the eigenvalues of $\mathbf{A}$) is independent of τ, since $\tau > 0$ only scales the eigenvalues without changing their sign. Therefore, this analysis predicts long-term behaviour based on the local linearisation of the one-step dynamics function F.

As shown in Fig. 6, for 1000 parameter configurations, all eigenvalues have negative real parts, with many close to -1. This arises from the structure of the Jacobian matrix. Since each neuron's update depends only on a subset of other neurons (Eqs. (4)–(6)), most diagonal elements of $\mathbf{J}$ are zero. In the linearised dynamics, $\mathbf{A} = \mathbf{J} - \mathbf{I}$, so these zeros become -1 on the diagonal of $\mathbf{A}$. Because neurons are weakly coupled to the rest of the network, these directions naturally produce eigenvalues near -1.

4 Homeostatic Plasticity: $\Delta\theta$ Preserving Output

Referring to the original experiment (Section 2.1), sensory deprivation causes axonal shrinkage, yet the circuit's output remains stable. We can use our model to investigate this phenomenon. Specifically, we can explore how parameters θ might vary while preserving the same output y, allowing us to identify which parameters may compensate for the experimentally observed structural changes.

To find parameter changes $\Delta\theta$ that preserve the output y, we need to solve $y(\theta + \Delta\theta) = y(\theta)$. Taking a Taylor expansion up to the quadratic term:

$$y(\theta_o+\Delta\theta) \approx\ y(\theta_o) + g^T\Delta\theta + \frac{1}{2}\Delta\theta^T\mathbf{H}\,\Delta\theta + \mathcal{O}(\|\Delta\theta\|^3) \quad (13)$$

where $g = \nabla_\theta y(\theta)$ is the gradient and $\mathbf{H} = \nabla^2_\theta y(\theta)$ the Hessian of the output. For changes in parameters to cause no changes in the output, i.e. for $y(\theta_o + \Delta\theta) \approx y(\theta_o)$:

$$g^T\Delta\theta + \frac{1}{2}\Delta\theta^T H\,\Delta\theta \approx 0 \quad (14)$$

Eq. (14) can be satisfied in two ways. One possibility is that both LHS terms are zero, i.e. $\Delta\theta$ is perpendicular to the gradient ($g^T\Delta\theta = 0$) and aligned with directions of eigenvectors with near-zero eigenvalues. This would produce a family of possible solutions. The other possibility is that the two terms exactly cancel each other out, which would produce a unique solution. Here we focus on the first scenario, as it aligns with the biological principle of homeostatic plasticity, whereby multiple weight configurations can compensate for perturbations, rather than relying on a single precisely tuned solution.

For a given initial parameter set (Section 3.4), we compute the Hessian of the mitral cell output y with respect to the parameters (θ) and examine its eigenvalues (Fig. 7).

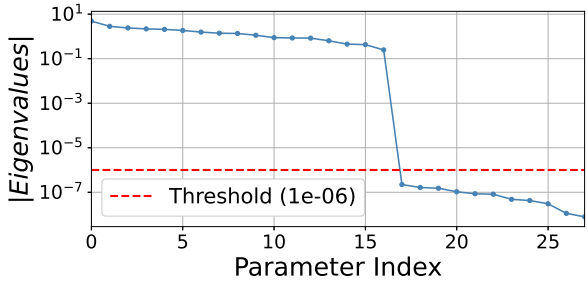


Figure 7: Distribution of eigenvalues ($|\lambda_i|$) of the Hessian matrix $\mathbf{H}$ for one parameter set.

Eigenvalues that are approximately zero ($|\lambda_i| < 10^{-6}$) indicate possible directions in parameter space along which synaptic

weights can vary without significantly affecting the OB output.

And from the corresponding eigenvectors, we choose those that are perpendicular to the gradient. As an example, Fig. 8 shows the components of one such eigenvector for the same set of initial parameters used in Fig. 7. The different components indicate how each parameter (synaptic strengths w, inhibitory feedback α, etc) are allowed to change without affecting the OB's output.

Currently, there are many possible stable parameter configurations that support directions in which parameters can change without affecting the network's output. These could be narrow down by selecting configurations that possess a larger number of near-zero eigenvalues.

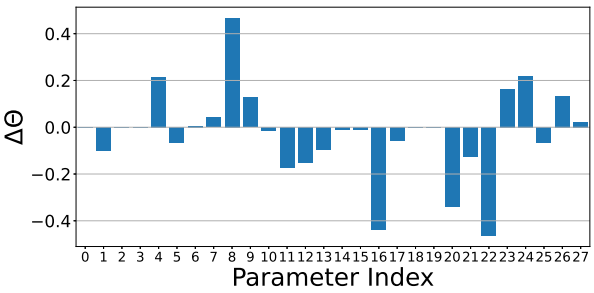


Figure 8: Eigenvector of a near-zero Hessian eigenvalue in parameter space.

The participation ratio (PPR) of the Hessian eigenvalues provides a measure of the effective number of directions in parameter space that influence the network output. For eigenvalues $\lambda_1, \ldots, \lambda_n$:

$$\mathrm{PPR} = \frac{\left(\sum_{i=1}^{n} |\lambda_i|\right)^2}{\sum_{i=1}^{n} \lambda_i^2} \tag{15}$$

Fig. 9 shows the PPR histogram for 1000 parameter sets. Smaller PPR values indicate sets with few dominant eigenvectors, and therefore, with many directions in which parameters can vary without affecting the output.

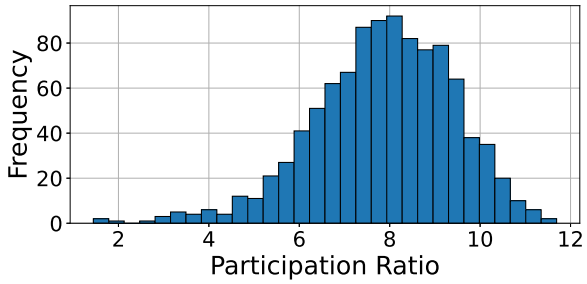


Figure 9: Histogram of PPR from Hessian eigenvalues of 1000 parameter sets.

5 Conclusion and Next Steps

We developed a simplified, biologically motivated model of the mouse olfactory bulb (OB) to investigate homeostatic plasticity observed in sensory deprivation experiments. The model shows that many stable synaptic configurations can produce the same OB firing rate. Using the Hessian, we identify parameter-space directions where synaptic and feedback changes do not alter network output.

Next, we will further constrain the parameter space using additional mathematical and biological criteria, including PPR-based filtering, excitatory–inhibitory balance, biologically plausible Naka–Rushton parameters, exclusion of silencing or saturating eigen-directions, and consistency with observed dopaminergic axon shrinkage. Whether solutions satisfying all constraints exist remains to be seen. If systematic synaptic patterns emerge, we will relate them to OB biology to inform future experiments.

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